

SIMATS ENGINEERING

ASSESSMENT TOOL 1

Constraint-Based Problem Solving

Course Name: Computer Networks

Course Code: CSA07

Course Outcome: CO5 — Design network topologies (star, mesh, bus, and hybrid) using networking devices, transmission media, and cabling standards

Assessment Weightage: 40%

QUESTIONS: 3

Total Marks:30

Annexure A

Constraint-Based Problem Solving

Code of Conduct

I, D. Anandpaul(192425207), certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:

Constraint-Based Problem Solving – Answers

Q1. Rural Branch Office — Connectivity, Devices, and Security Under Budget Constraints

Understanding the Constraints

This scenario presents four competing constraints that must all be satisfied together: no wired high-speed internet is available in the rural location, the office must reliably run cloud applications and video conferencing, secure communication with headquarters is mandatory, and the budget is limited. The design below addresses each constraint explicitly rather than optimising for just one at the expense of the others.

Recommended Internet Connectivity

Since fibre or DSL is unavailable, the most practical primary connection is a satellite broadband service (VSAT) or, where cellular coverage exists, a 4G/5G fixed-wireless connection. Satellite broadband providers increasingly offer business-grade plans with enough throughput for video conferencing and cloud applications, while modern satellite systems have significantly reduced latency compared to older geostationary services, making them usable for real-time collaboration.

To protect against the higher latency and occasional weather-related disruption of satellite links, a 4G/5G LTE router is recommended as a secondary backup WAN connection. A dual-WAN router configured for automatic failover ensures that if the primary satellite link degrades or drops, traffic seamlessly shifts to the cellular backup, preserving uninterrupted business operations without manual intervention.

Networking Devices Required

- Dual-WAN router / SD-WAN appliance — manages failover between satellite and LTE links automatically.
- Unified Threat Management (UTM) firewall — combines firewall, antivirus, and intrusion prevention in a single cost-effective device suited to a small branch.
- Managed switch (8–16 port) — connects local workstations, printers, and a Wi-Fi access point within the branch.
- VPN-capable endpoint (often built into the UTM firewall) — establishes an encrypted site-to-site tunnel back to headquarters.

Security Mechanisms

All traffic to headquarters should travel through an encrypted site-to-site VPN (IPsec or WireGuard), ensuring that sensitive business data cannot be intercepted over the public satellite or cellular link. The UTM firewall should also enforce basic content filtering and intrusion prevention to protect the branch from external threats, since a small office is unlikely to have dedicated on-site IT security staff to respond to incidents manually.

Bandwidth Considerations

Video conferencing typically requires at least 2–4 Mbps of stable bandwidth per simultaneous call, while general cloud application use adds further overhead; a mid-tier satellite or LTE plan in the 25–50 Mbps range comfortably supports a small branch office of a handful of users with room to spare. Quality-of-Service (QoS) rules should be configured on the router to prioritise video conferencing and VPN traffic over less time-sensitive downloads, ensuring call quality is preserved even when other applications are active.

Reliability Justification

The dual-WAN failover design directly addresses the requirement for uninterrupted business operations: satellite connectivity is not dependent on any local terrestrial infrastructure, and the cellular backup provides a second, independent path in case of satellite outages (for example, during severe weather). This redundancy is far more cost-effective than laying dedicated fibre to a remote rural site, which may not be feasible at any price in some locations.

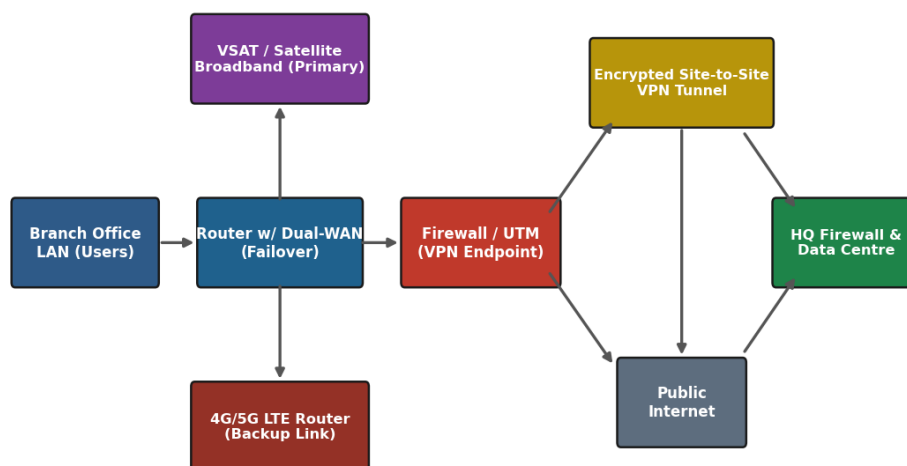
Cost Justification

Compared to alternatives such as leased-line connectivity (often impossible or prohibitively expensive in rural areas) or building dedicated infrastructure, satellite and cellular connectivity require no new civil works and can typically be provisioned within days at a predictable monthly cost. A single UTM appliance replacing several standalone security devices further reduces both capital and ongoing licensing costs, which matters for a budget-constrained branch office.

Scalability for the Future

As the branch grows, additional bandwidth can usually be added simply by upgrading the satellite or LTE plan tier, without replacing any hardware. The chosen UTM firewall and managed switch are typically available in a range of capacities, so if the office headcount grows, the same architecture scales by adding switch ports or a second access point rather than redesigning the network from scratch.

Rural Branch Office — Resilient Internet & VPN Connectivity



Q2. Campus-Wide Wireless Network for 3,000+ Students

Understanding the Requirements

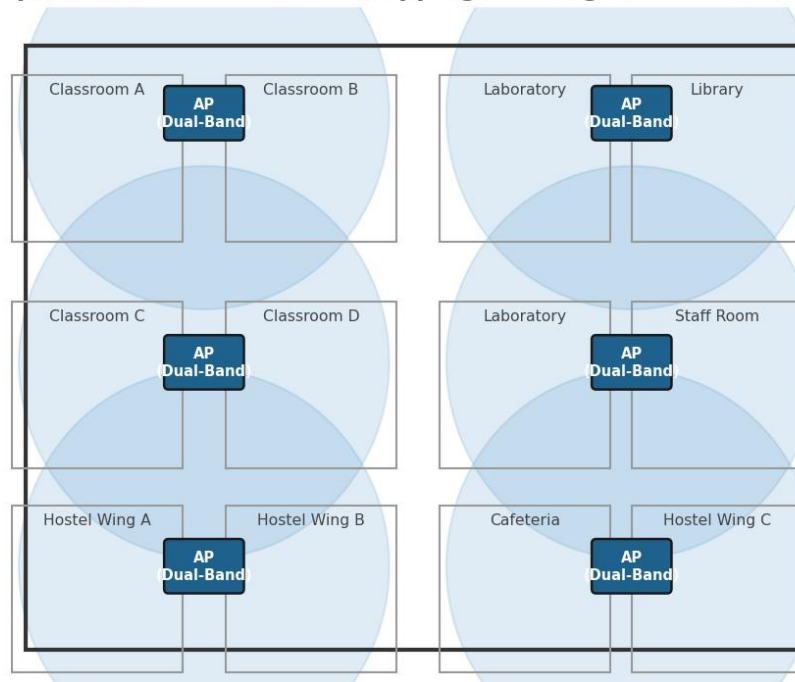
This scenario requires supporting a very large number of simultaneous wireless users across diverse spaces — classrooms, laboratories, libraries, and hostels — each with different usage patterns and device densities. The core tension is between coverage and capacity on one side and a limited budget for access points on the other, meaning every access point must be positioned and configured to serve as many users as efficiently as possible.

Access Point Placement Strategy

Rather than placing one access point per room, which would exceed the budget, access points should be positioned at the intersection of two to four adjacent rooms so that each device's coverage radius overlaps slightly with its neighbours. This overlapping cell design (similar in principle to cellular network planning) ensures there are no coverage gaps in corridors or shared walls, while minimising the total number of access points required to blanket the whole floor.

High-density areas such as libraries, lecture halls, and cafeterias should receive a higher concentration of access points than low-density areas such as hostel corridors, since these spaces have far more simultaneous active users per square metre. Ceiling-mounted access points are preferred over wall-mounted units where possible, as they provide more uniform, omnidirectional coverage across open study or classroom spaces.

Campus AP Placement — Overlapping Coverage, Minimal AP Count

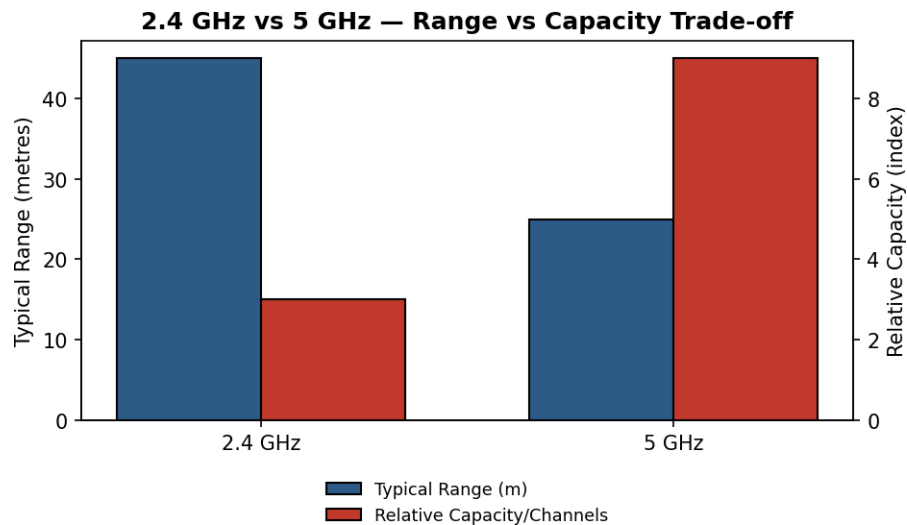


Frequency Band Selection

A dual-band deployment using both 2.4 GHz and 5 GHz is recommended, with client devices steered toward 5 GHz whenever signal strength allows. The 2.4 GHz band travels further and penetrates walls better, making it

useful for extending coverage into hostel rooms and corridors, but it offers fewer non-overlapping channels and is more prone to interference from other devices such as Bluetooth accessories and microwave ovens.

The 5 GHz band offers substantially more non-overlapping channels and higher throughput, which is critical in dense areas like libraries and lecture halls where dozens of devices may connect to a single access point simultaneously; its shorter range is actually an advantage here, since it reduces co-channel interference between neighbouring access points placed closer together in high-density zones. Modern access points supporting Wi-Fi 6 (802.11ax) further improve capacity in crowded areas through technologies such as OFDMA, which allows an access point to serve multiple clients in a single transmission window.



Authentication and Security

Given the scale of the university, WPA3-Enterprise authentication using 802.1X with a RADIUS server tied to the university's existing student and staff directory (such as Active Directory or LDAP) is recommended over a single shared password. This allows each student and staff member to authenticate with their own university credentials, enables per-user traffic accounting, and allows compromised or graduated accounts to be revoked individually without needing to change a network-wide password.

A separate guest network with a captive portal and rate-limited bandwidth should be provided for visitors, isolated from the main academic network through VLAN segmentation, so that guest traffic cannot reach sensitive internal systems such as student record servers or staff resources.

Balancing Coverage, Performance, Security, and Cost

The overlapping-cell placement strategy directly addresses coverage while keeping the access point count — and therefore cost — as low as possible for the required footprint. Assigning denser access point clusters and 5 GHz preference to high-traffic zones addresses performance where it matters most, without the expense of uniformly high-density deployment across low-traffic areas like hostel corridors.

WPA3-Enterprise with RADIUS integration addresses security without adding significant recurring cost, since it reuses infrastructure (the student/staff directory) that the university already operates for other purposes. Overall, this design concentrates spending on capacity in high-demand areas, accepts adequate (rather than

excessive) coverage in low-demand areas, and uses existing identity infrastructure for security — a balanced approach appropriate for a large, budget-constrained institution.

Q3. Secure Network Architecture for a Financial Institution

Understanding the Constraints

A financial institution must satisfy four constraints simultaneously: strong cybersecurity for highly sensitive customer and banking data, uninterrupted employee access to the systems needed for daily operations, strict regulatory compliance, and control over both performance impact and operational cost. The architecture below is built in layers, so that each security control addresses a specific risk without creating an unnecessary bottleneck elsewhere.

Network Segmentation Strategy

The network should be divided using VLANs into clearly separated zones: a DMZ (demilitarised zone) for externally facing services such as the bank's website and email gateway, and separate internal VLANs for teller/branch systems, loans and accounts, administration/HR, and core banking servers. This segmentation ensures that a compromise in one zone — for example, a phishing attack against an HR workstation — cannot directly reach core banking servers without passing through additional firewall and monitoring controls.

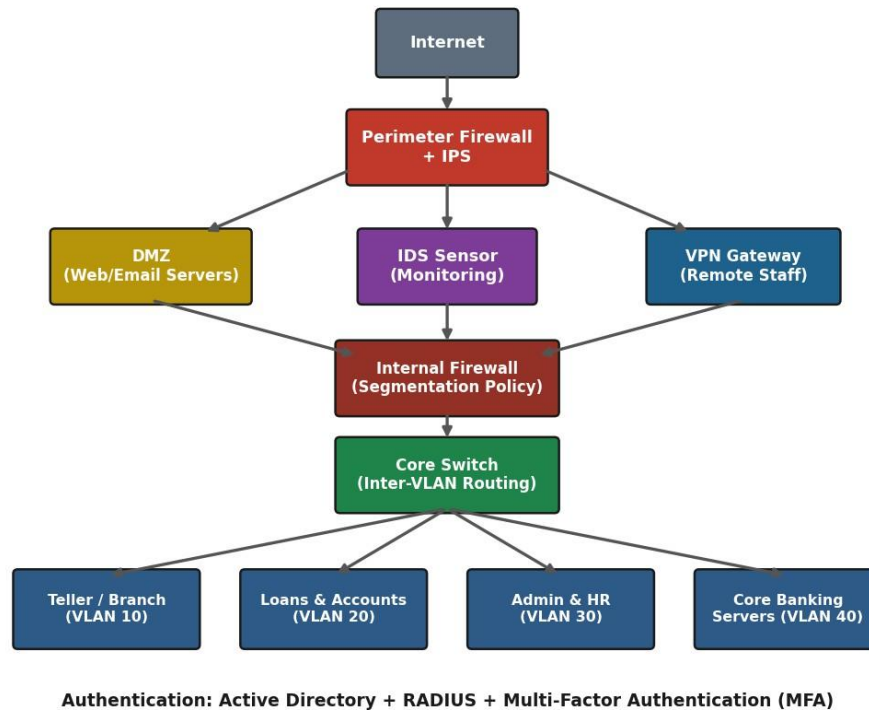
Placing core banking servers in their own highly restricted VLAN, separate from general staff VLANs, follows the principle of least privilege: only the specific applications and staff roles that genuinely need access to these systems are granted it, and all such access is explicitly logged for audit and compliance purposes.

Firewall Deployment

A perimeter firewall combined with an intrusion prevention system (IPS) sits at the boundary between the internet and the bank's network, inspecting and filtering all inbound and outbound traffic before it reaches the DMZ. A second, internal firewall then enforces segmentation policy between the DMZ, VPN gateway, and the various internal VLANs, ensuring that even traffic which has already passed the perimeter cannot move freely between internal zones without explicit policy permission.

This two-tier firewall design — perimeter plus internal — means that a single misconfiguration or vulnerability is far less likely to expose the entire network, since an attacker who breaches the outer perimeter still faces a second, independent barrier before reaching sensitive internal systems.

Financial Institution — Segmented Secure Network Architecture



Authentication Methods

Employee access should be governed by centralised Active Directory authentication integrated with a RADIUS server for network-level access control, combined with mandatory multi-factor authentication (MFA) for any access to core banking applications or remote access via VPN. This ensures that even if a password is compromised through phishing, an attacker cannot gain access without the second authentication factor, directly addressing regulatory requirements for strong authentication in the financial sector.

Intrusion Detection Mechanisms

An intrusion detection system (IDS) positioned to monitor traffic between the perimeter firewall and internal network provides continuous, passive analysis of traffic patterns, flagging anomalies such as unusual data transfer volumes or connection attempts to core banking servers from unexpected sources. Because the IDS operates passively (monitoring copies of traffic rather than sitting directly in the data path), it adds essentially no latency to normal operations, addressing the constraint that security controls must not significantly affect performance.

Compliance Considerations

Segmentation, MFA, and continuous monitoring together map directly onto common financial-sector compliance requirements (such as PCI DSS for card data and general banking regulatory frameworks), which typically mandate network segmentation of cardholder or customer data environments, strong authentication, and logging/monitoring of access to sensitive systems. Centralising logs from firewalls, the IDS, and authentication servers into a security information and event management (SIEM) system further supports compliance by providing an auditable trail of access and security events for regulators.

Cost Considerations

This architecture deliberately reuses infrastructure the bank already needs for other reasons — Active Directory for staff accounts, a core switch for routine inter-department traffic — rather than introducing entirely separate systems purely for security, which keeps additional capital expenditure focused on the firewall pair, IDS sensor, and MFA licensing. Passive IDS monitoring also avoids the higher cost of an inline intrusion prevention appliance at every internal boundary, since a single well-placed sensor can observe traffic across multiple internal links without needing to be deployed redundantly throughout the network.

Balancing Security, Performance, Compliance, and Budget

This layered design concentrates the strongest and most expensive controls (MFA, strict segmentation, dedicated core banking VLAN) around the most sensitive assets, while using more standard, lower-cost controls (perimeter firewall, passive IDS monitoring) to protect the broader network — an approach that satisfies strict security and compliance obligations without requiring uniformly expensive controls across every part of the network. Because the IDS is passive and the internal firewall enforces policy rather than performing deep computation on every packet, the design also avoids introducing a performance bottleneck into everyday employee workflows, keeping the bank's operations responsive while remaining secure and compliant.

Summary

Taken together, network segmentation, layered firewalls, strong authentication, and passive intrusion detection form a defence-in-depth architecture that addresses the financial institution's security and compliance obligations at each layer of the network, while deliberately avoiding controls that would either be prohibitively expensive or would slow down day-to-day employee access to the systems the bank depends on.